How Close Are We to the Edge? Measuring Our Distance from Vacuum Decay with DESI DR2

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Abstract
Every long-lived thing must stay distinct from nothing. That simple requirement motivates why

we live in 3 space + 1 time, why time has an arrow, and why empty space has energy. More importantly, it gives a number: any vacuum like ours must have a bounce action $S_E \geq 556$ to survive until today. We call the extra $\Delta = S_E - S_{floor}$ the distance to the edge. Using DESI DR2 data, we measure it for the first time.

1. Why this paper matters

For 100 years we have asked: Why 3 dimensions? Why does time flow? Why does vacuum have energy?

This paper does three things:

One idea instead of three. We start with one stability rule: a physical state, if it is to persist, must stay robustly distinguishable from absence and must not spontaneously fall back into it.

Formulated as a Lindblad evolution where the "absence" sector is repelling and the allowed sector is attracting, it naturally gives you separation (need space), structure (need stable orbits → 3 dimensions), and irreversibility (need 1 time arrow).[1][2]

A ruler. If our vacuum is metastable, it decays as $\Gamma/V = A \exp(-S_E)$ per volume. To have survived a Hubble time in a Hubble volume, we need $S_{floor} = \ln(A/H_0^4) \approx 556 + \ln(A/M_{Pl}^4)$. 556 is not magic — it is $4 \ln(M_{Pl}/H_0)$, the hierarchy between Planck and Hubble scale. We define $\Delta = S_E - S_{floor}$. If $\Delta \gg 1$, we are safe. If $\Delta \sim 0$-5, we live near-critical, like the Higgs vacuum which also sits at $S_E \sim 400$-600. If $\Delta < 0$, we shouldn't exist.[3][4][5]

A measurement. In the simplest decaying vacuum model, where vacuum energy leaks into

matter, the effective equation of state today is $w_0 = -1 + \Gamma/3H_0$. So DESI's measurement of w_0 is a measurement of Γ/H_0, and therefore of Δ. We implemented this in CLASS, ran Cobaya with DESI DR2 BAO + Planck + Pantheon+ SNe, and got:

$\Gamma/H_0 < 0.12$ (95% CL)

$\Delta \sim 0.3 - 2.5$ (68% CL), $\Delta > 0.1$ (95%)[10][11][14][15][16][6][7][8][9]

Data are consistent with living close to the edge, Δ of order one, not 100.

2. Extra dimensions?

If there are n extra dimensions with KK tower, they lower S_E by $\Delta S_E \sim -n \ln(M^*/m_{KK})$. If we are already near the edge, we cannot afford much lowering. So TeV-scale large extra dimensions with $n \geq 2$ are disfavored if near-criticality holds.[12][13]

3. What this is NOT

- Not claiming we derived S_floor from first principles. We motivate it.
- Not claiming we proved vacuum must decay this way. We work conditionally in decaying-Λ class.
- Not first to bound Γ — several groups bounded decaying DE with DESI. We are first to frame it as distance Δ to survival floor.

Data Availability

All CLASS patches, Cobaya yaml files, MCMC chains and plotting scripts are publicly available at Zenodo and GitHub.[17]

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